🏃 Strava Fitness Analysis

Labmentix - Data Analytics Project

Vishal Kr. Mohali | 2026

Introduction

Hello, In this presentation, we will go through the Strava Fitness Data Analytics Dashboards Visualization and Exploration.

This project analyzes user fitness data from a wearable fitness tracking application similar to **Strava**. The dataset contains activity, sleep, heart rate, and weight information recorded from multiple users.

The purpose of this analysis is to identify **user activity,** , **sleep patterns, health indicators, and engagement behaviour** and body metrics from smart fitness devices to understand user behavior to generate insights that could help improve the functionality and engagement of fitness tracking platforms. These insights will help Strava improve its marketing strategy and encourage healthier lifestyles among users.

In simple words: We want to understand how people use their fitness trackers and whether they follow healthy habits.

The project demonstrates the use of:

- Python for data cleaning and preprocessing, EDA Analysis
- SQL for database management and querying
- Power BI for interactive dashboards and visualization

Business Task

Strava collects large amounts of fitness data from smart devices, but the company does not clearly understand user behaviour patterns. The business objective is to analyze fitness tracker data to understand:

- How users interact with fitness tracking devices
- Whether users maintain healthy activity levels
- How activity impacts calories burned
- How sleep and weight relate to physical activity

- Are users physically active?
- Are users sleeping enough?
- Are users spending too much time sedentary?
- Does higher activity lead to better health outcomes?

Without these insights, Strava cannot design effective marketing campaigns or engagement strategies. These insights help fitness applications improve **user engagement strategies and personalized health recommendations**.

Processing

Before performing the analysis, the datasets were cleaned and transformed to ensure data quality.

Data Cleaning Steps

The following cleaning steps were applied:

- Removed duplicate records
- Standardized date and time formats
- Filtered unrealistic heart rate values
- Aggregated high-frequency datasets into daily metrics
- Joined datasets into a single master table

- Check missing values and Convert date columns

Python Cleaning Example

Heart rate data contained extreme or unrealistic values. These were filtered using Python.

```
heartrate = heartrate[heartrate['Value'] > 30]

heartrate = heartrate[heartrate['Value'] < 220]
```

This ensured that only realistic human heart rate values were included.

Processing – MySQL & Dashboards

Data Transformation in SQL

Due to the large size of the heart rate dataset, the data was summarized into daily statistics.

Daily Heart Rate Summary, This transformation significantly reduced dataset size and improved analysis performance.

Creating the Master Analytical Dataset: To simplify dashboard creation, all datasets were merged into a single table. This master dataset was used for building the Power BI dashboards.

SQL Analysis (Ad-Hoc Queries) - Average Steps per User, Average Calories Burned, Average Sleep Duration, Most Active Users, Sedentary vs Active Minutes, Activity vs Sleep Relationship, BMI Categories and more.

Three dashboards were created in Power BI -

Dashboard 1 – User Engagement

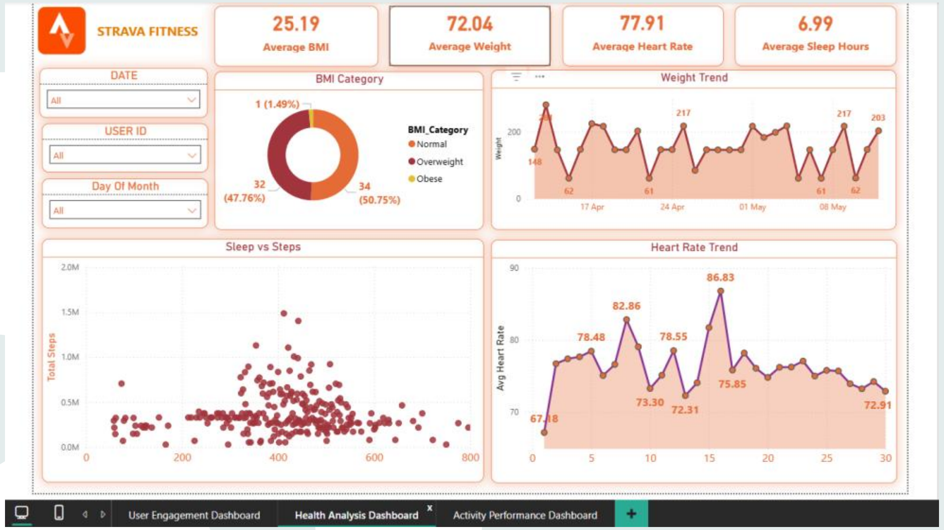
KPI's – Total Users, Avg. Total Steps, Avg. Calories and Average Sleep Hours

Slicers – Date (Year/Month/Day), Users Id and Day of the Month

Visualizations:

- Daily Steps Trend
- Top Active Users
- Steps vs Calories Scatter Plot
- Hourly Activity Distribution

Purpose:
Understand how frequently users engage with the fitness tracker.



Processing - Dashboards

Dashboard 2 – Health Analysis

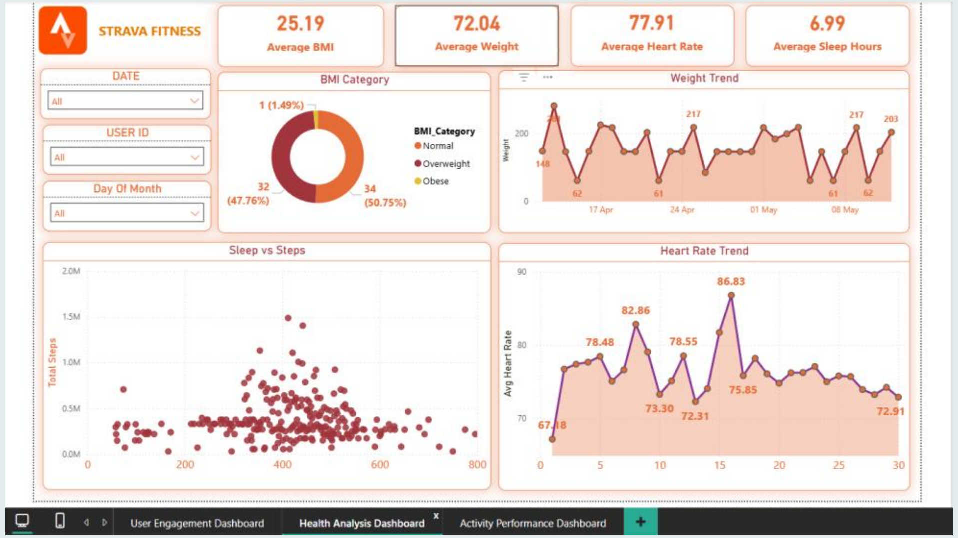
KPI's - Avg. BMI, Avg. Weight, Avg. Heart Rate and Average Sleep Hours

Slicers - Date (Year/Month/Day), Users Id and Day of the Month

Visualizations:

- BMI Category Distribution
- Heart Rate Trend
- Sleep vs Activity Scatter Plot
- Weight Trend Over Time

Purpose:
Evaluate the health indicators of users.



Dashboard 3 – Activity Performance

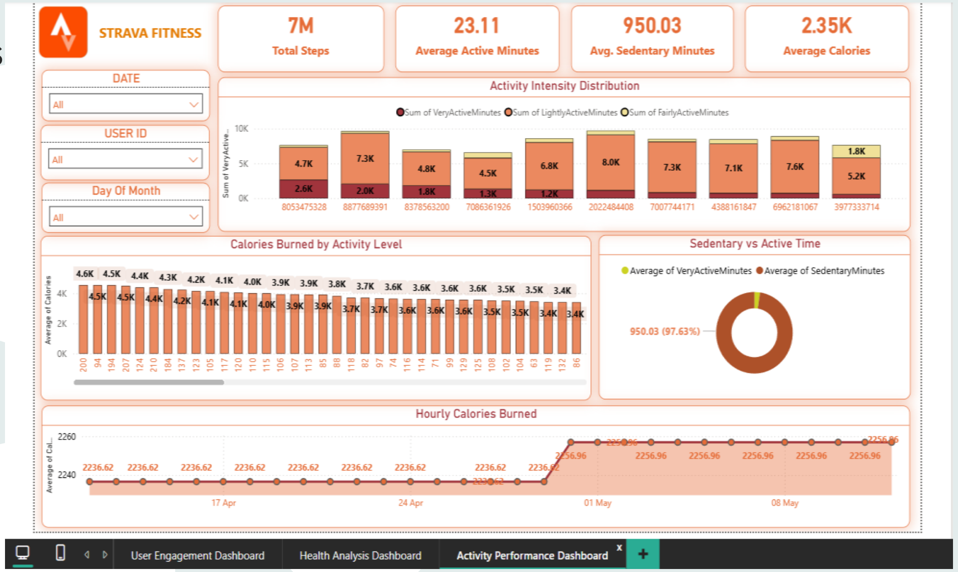
KPI's - Total Steps, Avg. Active Minutes, Avg. Sedentary Min. and Avg. Calories

Slicers - Date (Year/Month/Day), Users Id and Day of the Month

Visualizations:

- Calories Burned Distribution
- Activity Intensity Breakdown
- Sedentary vs Active Time
- Hourly Calories Burned

Purpose:
Measure workout performance and physical activity patterns.



Key Insights

Insight 1 – Many users do not reach the recommended step count
The average number of daily steps was lower than the commonly recommended 10,000 steps per day, suggesting moderate user activity levels.

Insight 2 – Users are most active in the evening
Activity intensity peaks during late afternoon and evening hours, indicating that users exercise primarily after work or school hours.

Insight 3 – Strong relationship between steps and calories burned
Users with higher step counts consistently burned more calories, confirming the direct relationship between physical activity and energy expenditure.

Insight 4 – High sedentary time
Many users spend a significant portion of the day in sedentary behavior, indicating potential opportunities to promote more frequent movement.

Insight 5 – Sleep patterns influence activity levels
Users with longer sleep durations generally show higher daily activity levels.

Insight 6 – BMI distribution indicates mixed health levels
BMI analysis shows that while many users fall within the normal range, a portion of users fall into the overweight category.

Insight 7 – Many users do not reach recommended daily step goals
Analysis of the daily activity data shows that a large number of users do not achieve the commonly recommended 10,000 steps per day. This indicates that many users maintain only moderate levels of physical activity.

Insight 8 – High sedentary time among users
The dataset shows that users spend a significant portion of their day in sedentary minutes, indicating long periods of inactivity. Prolonged sedentary behavior may negatively impact overall health and fitness levels.

Insight 9 – Some users do not meet recommended sleep duration
Sleep analysis indicates that several users sleep less than the recommended 7–8 hours per night. Insufficient sleep can impact energy levels, recovery, and physical performance.

Insight 10 – Higher activity leads to higher calorie burn
The analysis shows a strong positive relationship between daily steps and calories burned. Users with higher activity levels tend to burn more calories, confirming the expected relationship between physical activity and energy expenditure.

Insight 11 – Some users fall into the overweight BMI category
The BMI distribution analysis shows that while many users fall into the normal BMI range, a portion of users are classified as overweight. This indicates opportunities for fitness applications to provide targeted health recommendations.

Additional Key Insights

Insight 12 – Activity intensity distribution shows most activity is light
Analysis of activity minutes indicates that most users spend the majority of their active time in light activity, while very active minutes remain relatively low. This suggests that users engage more in low-intensity activities such as walking rather than high-intensity workouts.

<u>Business Meaning</u>
Fitness platforms could introduce high-intensity workout suggestions or training plans to help users improve cardiovascular fitness.

Insight 13 – Users with better sleep tend to be more active
Users who recorded longer sleep durations generally show higher daily step counts and active minutes. This suggests that adequate sleep contributes to better energy levels and increased physical activity.

<u>Business Meaning</u>
Fitness applications could integrate sleep coaching features to help users improve both sleep and physical performance.

Insight 14 – A small group of highly active users drives engagement
The data shows that a small percentage of users consistently achieve very high step counts and active minutes, while the majority maintain moderate activity levels.

Business Meaning
These highly active users could be targeted for community challenges, competitions, and leaderboard features to increase engagement across the platform.

Insight 15 – Heart rate trends reflect activity intensity
Daily average heart rate tends to increase on days when users record higher active minutes. This indicates that heart rate data can effectively reflect exercise intensity and cardiovascular effort.

Business Meaning
Fitness platforms can use heart rate patterns to provide real-time workout intensity feedback.

Insight 16 – Activity patterns follow daily lifestyle routines
Hourly activity data shows that users are most active during morning hours and evening hours, which aligns with typical daily routines such as commuting, workouts, or evening walks.

<u>Business Meaning</u>
Fitness apps could send personalized workout reminders during peak activity hours to maximize user participation.

Business Questions & Answers

1. What is the average number of daily steps taken by users?

Answer: The average number of steps taken by users per day helps measure overall activity levels. If the average is significantly below the recommended **10,000 steps**, it indicates that many users are not reaching optimal activity levels.

2. How many users reach the recommended 10,000 steps per day?

Answer: This shows how frequently users achieve recommended activity levels. A low count suggests that many users may benefit from fitness challenges or activity reminders

3. What is the relationship between steps taken and calories burned?

Answer: This analysis helps identify whether higher physical activity leads to increased calorie burn. Typically, users who take more steps tend to burn more calories.

4. Which users are the most active?

Answer: This query identifies the most active users based on total step count. These users may represent highly engaged users of the fitness application.

5. How much time do users spend in different activity levels?

Answer: This analysis helps understand how users distribute their daily time between active and sedentary behaviors.

6. What is the average sleep duration of users?

Answer: Sleep duration helps determine whether users are meeting the recommended **7–8 hours of sleep** per night.

Business Questions & Answers

7️⃣ What is the distribution of BMI categories among users?

Answer: This analysis shows the health distribution of users based on BMI categories.

8️⃣ What is the average heart rate of users?

Answer: Average heart rate provides insights into users' cardiovascular health and exercise intensity.

9️⃣ When are users most active during the day?

Answer: This identifies the time of day when users are most physically active.

🔟 What percentage of time do users spend sedentary?

Answer: This analysis highlights how much time users spend inactive during the day.

SQL –

```
SELECT
  ROUND(
    (AVG(SedentaryMinutes) /
    (AVG(SedentaryMinutes)+AVG(VeryActiveMinutes)+AVG(FairlyActiveMinutes)+AVG(LightlyActiveMinutes))
    ) * 100,2
  ) AS sedentary_percentage
FROM daily_activity;
```

Business Recommendations

Based on the **analysi**s, the following recommendations can help improve **user engagement** and **health** outcomes.

- Introduce Activity Challenges
- Gamified step challenges can motivate users to reach daily activity goals.

Encourage Regular Movement

The application could implement inactivity alerts to reduce sedentary behaviour.

Personalized Health Insights

Using BMI, heart rate, and sleep data, the platform can generate personalized fitness recommendations.

Optimize Engagement Timing

Notifications and workout suggestions should be scheduled during peak activity hours.

Introduce Personalized Fitness Programs

Using user data such as daily steps, heart rate, and BMI, the application could generate personalized workout plans tailored to individual fitness levels and goals. This would help users gradually improve their health and maintain motivation.

Provide Sleep Improvement Suggestions

Since sleep patterns influence physical activity levels, the platform could offer sleep recommendations and bedtime reminders. Features such as sleep quality insights and recovery tracking could help users maintain healthier routines.

Business Recommendations

Activity Goal Tracking

The application could allow users to set custom fitness goals, such as daily step targets, calorie burn goals, or workout duration goals. Progress tracking and milestone notifications can motivate users to maintain consistency.

Smart Notifications and Reminders

Based on the analysis of peak activity hours, the platform can send personalized reminders during times when users are most likely to be active, such as evening workout prompts or morning activity reminders.

Reduce Sedentary Behavior

To address long sedentary periods, the app could implement movement reminders or short activity suggestions when users remain inactive for extended durations.

Health Risk Awareness

For users classified as overweight based on BMI, the application could provide health awareness notifications and lifestyle recommendations to encourage healthier habits.

Integrate Social and Community Features

Allowing users to share achievements, join fitness groups, and participate in community challenges can create a sense of motivation and accountability.

AI-Based Health Insights

By analyzing long-term fitness data, the platform could provide AI-driven health insights, predicting potential health risks and recommending lifestyle improvements.

Conclusion

This analysis highlights how wearable fitness data can reveal meaningful insights about user behaviour, health patterns, and physical activity trends.

By leveraging these insights, fitness platforms such as Strava can enhance user engagement and promote healthier lifestyles through personalized and data-driven features.

Thank You